

Linear Algebra — Week 01

Answer Key & Misconception Notes

EXAMPLE 1 — Answer

$$z_1 + z_2 = 2 + 2i; z_1 - z_2 = 4 - 6i; z_1 \times z_2 = 5 + 14i$$

Common misconception: Students often forget $i^2 = -1$ when expanding the product, leaving a bdi^2 term uncombined.

EXAMPLE 2 — Answer

$$|z| = 13$$

Common misconception: Confusing modulus with $|\operatorname{Re}(z)|$ alone (i.e. forgetting the imaginary part).

EXAMPLE 3 — Answer

$$z^{-1} = \frac{2}{13} - \frac{3}{13}i$$

Common misconception: Sign error: using $+\frac{b}{a^2 + b^2}$ instead of $-\frac{b}{a^2 + b^2}$ in the inverse formula.

EXAMPLE 4 — Answer

$$\frac{2+i}{1-i} = \frac{(2+i)(1+i)}{(1-i)(1+i)} = \frac{1+3i}{2} = \frac{1}{2} + \frac{3}{2}i$$

Common misconception: Forgetting to multiply *both* numerator and denominator by the conjugate.

EXAMPLE 5 — Answer

$$\bar{z} = -4 - 7i; z + \bar{z} = -8 \text{ (real)}$$

Common misconception: Thinking the conjugate flips the sign of a as well as b .

EXAMPLE 6 — Answer

—

Common misconception: Mixing up A3 (identity) with A4 (inverse), or forgetting A3 requires *both* an additive **and** multiplicative identity.

EXAMPLE 7 — Answer

—

Common misconception: Believing “subtraction is fine because $2 - 3$ makes sense” — misses that the axiom requires commutativity for *all* pairs, not that the operation is merely defined.

EXAMPLE 8 — Answer

Not a field — A7 fails (no multiplicative inverse for most integers, e.g. 2)

Common misconception: Assuming closure alone is enough to be a field; forgetting inverses must also lie in the set.

EXAMPLE 9 — Answer

$$y = \vec{0}$$

Common misconception: Trying to “cancel” x from both sides informally instead of adding the inverse and citing A1/A2/A3/A4 explicitly.

EXAMPLE 10 — Answer

Uniqueness proof mirrors zero-vector proof, swapping roles

Common misconception: Forgetting to use commutativity (A2) as the pivot step.

EXAMPLE 11 — Answer

$$-(-x) = x$$

Common misconception: Treating $-x$ as a new independent object rather than $(-1)x$ from Example 5, missing the axiomatic route entirely.

EXAMPLE 12 — Answer

$$2 \boxdot x = (1 + 1) \boxdot x = 1 \boxdot x \boxplus 1 \boxdot x = x \boxplus x \text{ (uses A6, then A8)}$$

Common misconception: Assuming $2x$ “obviously” means $x + x$ without justifying it from the axioms — the whole point of this unit is that nothing is obvious until proven.